ML PRACTICAL 3

Diabetes Prediction using K-Nearest Neighbors (KNN)

Aim

To implement the K-Nearest Neighbors algorithm to predict diabetes using the diabetes dataset and evaluate its performance.

Theory

Diabetes prediction is a supervised machine learning classification task. In this practical, the **diabetes.csv** dataset is used to predict whether a person has diabetes. The **Outcome** column is the target variable, where 1 represents diabetic and 0 represents non-diabetic. The remaining columns are used as input features.

Data Preprocessing

The dataset is divided into 82% training data and 18% testing data using train_test_split. Stratification maintains the class proportions in both sets. StandardScaler is used to standardize the input features so that variables with larger numerical ranges do not dominate distance calculations.

K-Nearest Neighbors (KNN)

KNN is a supervised learning algorithm used for classification. It calculates the distance between a new data point and the training points, selects the nearest K neighbors, and assigns the majority class among them. In this practical, K is set to 7.

For distance-based KNN, feature scaling is important because the algorithm relies on distances between data points.

Model Evaluation

The trained model predicts the classes of the test data. Its performance is evaluated using:

- **Confusion Matrix:** Summarizes correct and incorrect predictions for both classes.
- **Accuracy:** Proportion of all predictions that are correct.
- **Error Rate:** Proportion of incorrect predictions (1 – accuracy).
- **Precision:** Proportion of predicted positive cases that are actually positive.
- **Recall:** Proportion of actual positive cases correctly identified.

Result

The KNN algorithm is implemented to predict diabetes using the given dataset. The model's performance is measured using a confusion matrix, accuracy, error rate, precision and recall.

Conclusion

This practical demonstrates how KNN can be used for diabetes classification. Feature standardization supports distance-based prediction, while evaluation metrics help assess the model's classification performance.

Theory prepared from the uploaded ML_Prac_3 notebook. Actual metric values should be taken from the executed notebook output.